THE SIMPLEX METHOD in TABULAR FORM

LINEAR PROGRAM RECAP

WYNDOR GLASS CO. PROBLEM (augmented - standard - form)

$$\max\ Z = 3x_1 + 5x_2 \qquad (0)$$

$$\text{st} \quad x_1 + x_3 = 4 \qquad (1)$$

$$2x_2 + x_4 = 12 \qquad (2)$$

$$3x_1 + 2x_2 + x_5 = 18 \qquad (3)$$

$$x_j \geq 0 \quad j = 1 \ldots 5$$

TABULAR FORM

BASIC VAR	EQ	Z	x_1	x_2	x_3	x_4	x_5	RHS	
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	12/2 = 6
x_5	(3)	0	3	2	0	0	1	18	18/2 = 9

Easy to visualize!

- Still negative values in row (0)
  ⇒ not yet optimal
- x_2 ENTERING BASIC VARIABLE (-5 most negative)
- x_4 LEAVING BASIC VARIABLE (MINIMUM RATIO TEST)
- x_2 PIVOT COLUMN
  x_4 PIVOT ROW

REMARK: we need to consider only the rows with strictly positive (>0) coeff in the pivot column

⇒ the goal is to restore the 0-1 pattern in the pivot column (1 in the pivot row, 0 otherwise)
⇒ GAUSSIAN ELIMINATION

$$R_2 \to R_2/2$$

$$R_0 \to R_0 + \frac{5}{2}R_2$$

$$R_3 \to R_3 - R_2$$

ITER	BASIC VAR	EQ	Z	X_1	X_2	X_3	X_4	X_5	RHS	
0	Z	(0)	1	-3	-5	0	0	0	0	
	X_3	(1)	0	1	0	1	0	0	4	
	X_4	(2)	0	0	2	0	1	0	12	
	X_5	(3)	0	3	2	0	0	1	18	
1	Z	(0)	1	-3	0	0	5/2	0	30	
	X_3	(1)	0	1	0	1	0	0	4	4/1 = 4
	X_2	(2)	0	0	1	0	1/2	0	6	
	X_5	(3)	0	3	0	0	-1	1	6	6/3 = 2

ENTERING BASIC VARIABLE

PIVOT COLUMN OF ITER 0 HAS NOW 0-1 PATTERN WHILE OTHER COLUMNS HAVE CHANGED ACCORDINGLY

START AGAIN:

- still negative values in row (0) => not yet optimal
- X_1 ENTERING BASIC VAR (-3 most negative)
- X_5 LEAVING BASIC VAR (MINIMUM RATIO TEST)

=> GAUSSIAN ELIMINATION to obtain 0-1 pattern

ITER	BASIC VAR	EQ	Z	X_1	X_2	X_3	X_4	X_5	RHS
2	Z	(0)	1	0	0	0	3/2	1	36
	X_3	(1)	0	0	0	1	1/3	-1/3	2
	X_2	(2)	0	0	1	0	1/2	0	6
	X_1	(3)	0	1	0	0	-1/3	1/3	2

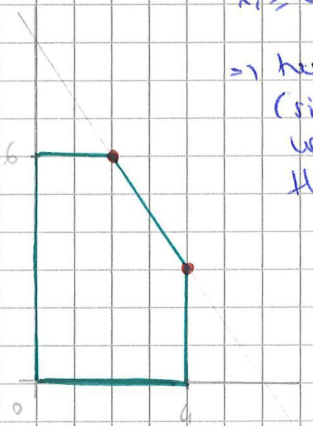
NO NEGATIVE COEFF!

OPTIMAL SOL FOUND !!

COMPLICATIONS

- MULTIPLE OPTIMAL SOLUTIONS
  In the Wyndor Glass Co. example, change the coefficient of x_2 in the cost function to 2 instead of 5

$$\max Z = 3x_1 + 2x_2$$
$$x_1 \leq 4$$
$$2x_2 \leq 12$$
$$3x_1 + 2x_2 \leq 18$$
$$x_1 \geq 0 \quad x_2 \geq 0$$



=> two optimal CPF sol (i.e. BFS)
(simply continue pivoting, choosing non-basic variable with a 0 in the Z row as the entering basic variable)

x_1 entering (most negative)
x_3 leaving (MIN ratio)

ITER	BASIC VAR	EQ	Z	x_1	x_2	x_3	x_4	x_5	RHS	
0	Z	(0)	1	-3	-2	0	0	0	0	
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	3	2	0	0	1	18	
1	Z	(0)	1	0	-2	3	0	0	12	
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	0	2	-3	0	1	6	
2	Z	(0)	1	0	0	0 !	0	1	18	OPTIMAL SOL FOUND! BUT...
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	0	3	1	-1	6	
	x_2	(3)	0	0	1	-3/2	0	1/2	3	
EXTRA	Z	(0)	1	0	0	0	0 !	1	18	STILL OPTIMAL!
	x_1	(1)	0	1	0	0	-1/3	1/3	2	
	x_3	(2)	0	0	0	1	1/3	-1/3	2	
	x_2	(3)	0	0	1	0	1/2	0	6	

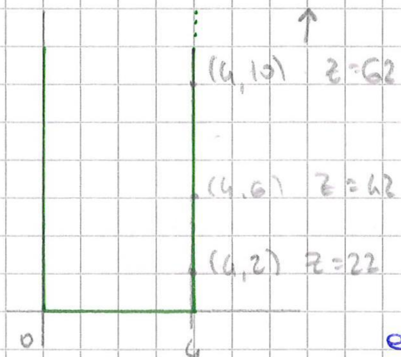
- modern LP solvers provide multiple CPF/BFS optima
- How to list all of the optimal CPF/BFS is a topic of research by itself!
- very similar to finding all the CPF/BFS of the feasible region
- The list can be very helpful to characterize the ENTIRE SPACE of (optimal) solutions

=> not part of this course!! :)

- **NO LEAVING BASIC VARIABLE**

Consider the Wyndor Glass Co. problem with only the first constraint

$$\max Z = 3x_1 + 5x_2$$
$$x_1 \leq 4$$
$$x_1 \geq 0 \quad x_2 \geq 0$$



BASIC VAR	EQ	Z	x_1	x_2	x_3	RHS
Z	(0)	1	-3	-5	0	0
x_3	(1)	0	1	0	1	4

shouldn't we avoid rows with coeff ≤ 0?

How can we compute the MINIMUM RATIO TEST?

=> we can increase x_2 arbitrarily! (still feasible)

=> UNBOUNDED OPTIMAL SOLUTION
UNBOUNDED OBJECTIVE FUNCTION

- **DEGENERACY**

- TIE for most negative value on row (0)
  i.e., TIE for entering basic variable
  => multiple non-basic variables with the same potential

- TIE for minimum ratio test
  i.e., TIE for leaving basic variable

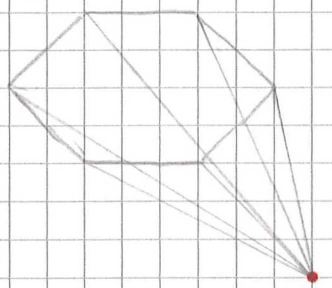
=> How do we solve this? ARBITRARILY!

HONEST ANSWER: let's address this with math!

- A BFS is **DEGENERATE** if one or more of its basic variables is zero,
  => DEGENERATE BASIC VARIABLES

Recall the concept of ADJACENCY: many definitions say that a CPF solution can have **at most** n adjacent CPF solutions (relying on the fact that in a n-dimensional space a CPF solution requires n constraints to be tight (i.e., met with equality) BUT...

EXAMPLE:



THE RED CPF is ADJACENT to 6 other CPF! (6 > 3)

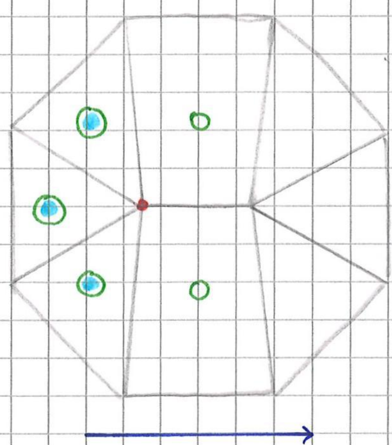
Any 3 of the 6 long faces of the diamond define a point

If we think of the slack variables of the three faces we choose as non-basic variables, then the slack variables of the other faces, incident at the point are basic but are also zeros! This basic variables are **DEGENERATE** and the CPF/BFS is **DEGENERATE**
(they have to be zero because a basis can only have 3 elements)

=> DEGENERATE CPF = more than n constraints are tight at that point

PROBLEM:
- the simplex method might be forced to go from one basis to the next without increasing the value of the objective function
  => **STALLING**
- the simplex method might be stuck in an eternal loop
  => **CYCLING**

STALLING: sometimes inevitable
sometimes more complicated than other times

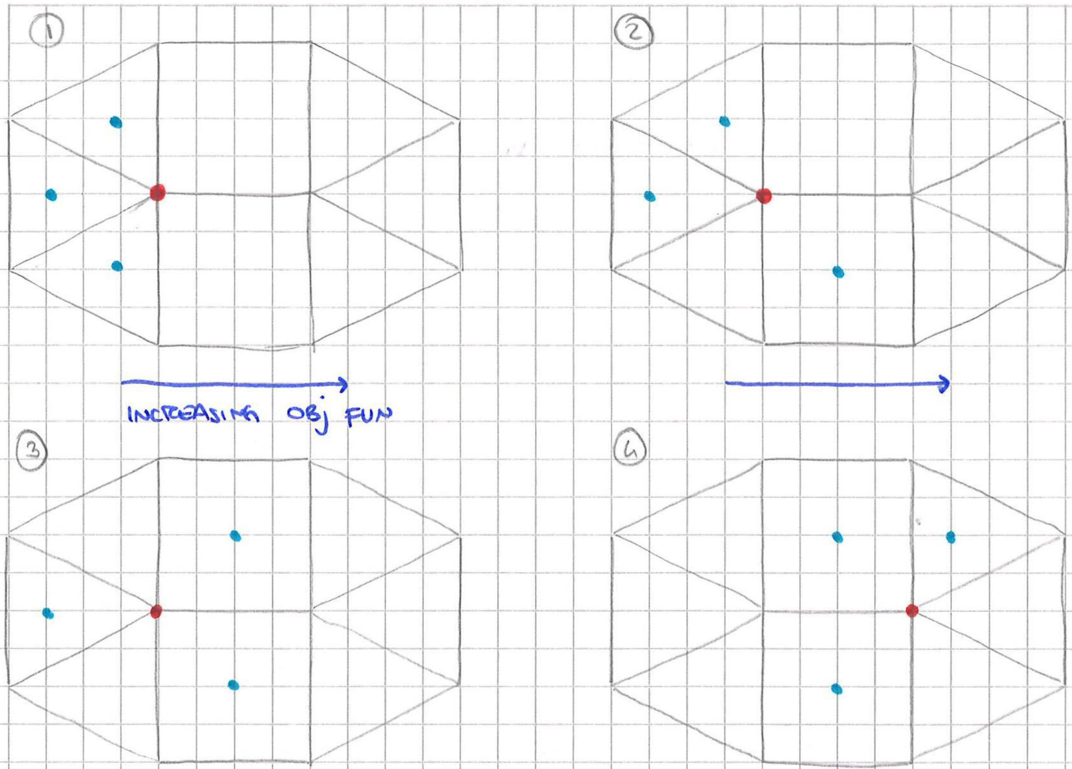


DIRECTION OF INCREASING OBJECTIVE FUNCTION

(horizontal line sticking out)

BFS - DEGENERATE!
- ● basic variable in row
- ○ slack variable = 0

We can move in the direction of increase with the basic variables but...



The objective function was improved only after 3 steps!
3 PIVOTS!

Algebrically you can see it coming:

BASIC VAR	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS	
*			-7			0	0	*	
x_3			2			1	0	6	6/2 = 3
*						0	0	*	
x_4			4			1	0	12	12/4 = 3
*								*	

We decide x_4 is the LEAVING VARIABLE

			0			0	0	
						0	0	
x_3			0			1	-1/2	$\boxed{0}$
						0	0	
x_2			1			1/4	0	3

x_3 is still in the basis but with value 0! (as expected)

Suppose x_1 is the next ENTERING VARIABLE

		-4	0			0	0		
*						0	0		
x_3		6	0			1	-1/2	0	0/6 cannot loose the minimum ratio test
*						0	0		
x_2		2	1			1/4	0	3	
*						0	0		

When x_3 leaves the basis, there will be no increase in the objective function!

COMMENTS:
- There are techniques for dealing with stalling but cannot be avoided in all cases
- Usually not a problem BUT
- if the mathematical structure of the LP is vulnerable to degeneracy, it can be a problem
  - depends on the specific problem
- if the simplex solver is stalling repeatedly and badly, you might want to switch to an interior point method! (not in this course)

CYCLING:
- In theory it is possible to loop around BFS
- since the objective function does not increase, there is a chance to return in a basis that we have seen before
  => there exists pivot rules which ensure that the simplex method will terminate

  BLAND'S RULE: pick the improving variable with smallest index and if there are several possibilities for the leaving variable also take the one with the smallest index

  => too slow in practice even if it guarantees that the simplex method will terminate

EXERCIZE: 4.4-5
4.4-6
4.4-8
4.5-4
4.5-8

PROPERTIES of CPF SOL

1. If there is EXACTLY ONE OPTIMAL SOL then it MUST BE a CPF SOL

   GEOMETRICAL PROOF: for problems with ONE OPTIMAL SOL, it is always possible to KEEP RISING the OBJECTIVE FUNCTION LINE (HYPERPLANE in HIGHER DIM) until it touches just ONE POINT at a CORNER FEASIBLE REGION (the OPTIMAL SOL).

2. If there are MULTIPLE OPTIMAL SOL (and a BOUNDED FEASIBLE REGION) then at least TWO MUST BE ADJACENT CPF SOL

=> According to these two PROPERTIES, we can check only CPF solutions to find the OPTIMAL SOL

3. There are ONLY a FINITE NUMBER of CPF SOL

   INTUITION: CPF sol are SIMULTANEOUS SOL of a SYSTEM of n out of n+m constraints. The number of POSSIBLE COMBINATIONS of the CONSTRAINTS is an UPPER BOUND on the number of SOLUTIONS

=> POSSIBLE SOL can be OBTAINED BY ENUMERATION

4. If a CPF SOL has ~~no~~ NO ADJACENT CPF SOL that are BETTER then there are NO BETTER CPF SOL ANYWHERE

   => SUCH a CPF SOL is guaranteed to be OPTIMAL (assuming the problem has at least one optimal sol - and the problem has feasible solutions and a bounded feasible region)

   It follows from the fact that the FEASIBLE ~~S~~ SET of an LP is CONVEX.